

FILE : CH10EG1

```
L=['q','w','e','r','t','y']
length=len(L)
for a in range(length):
    print("at index",a,"and",(a-length),"element:",L[a])
```

FILE : CH10EG1A

```
#1
print()
print("#1")
a=[10,12,14]
a+="abc"           # list can be added to a string when it is a
accumulator(+=)
print(a)           ##### "abc" is treated as a list and not as a
string
```

```
#2
print()
print("#2")
a=[10,12,14]
#a=a+"abc"         list can't be added to a string
print(a)
```

```
#3
print()
print("#3")
a=[10,12,14,20,22,24,30,32,34]
print(a[3:-3])
```

```
#4
print()
print("#4")
L=["one","two","three","four"]
print(L)
L[0:1]=[4,5]
print(L)
L[0:2]=[0,1]
print(L)
```

```
L[0:3]=[0,1]
print(L)
```

```
#5
print()
print("#5")
L[0:-2]=[345]
print(L)
```

```
#6
print()
print("#6")
a=[1,2,3]
a1=[1,2,3]
b=a
a[1]=5
print(a)
print(b)
a1[1]=5
b=a1
b=list(a1)
print(b)
```

FILE : CH10EG2

```
lst=[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20]
slc1=lst[5:15:2]
slc2=lst[::4]
sum=avg=0
print("slice1")
for a in slc1:
    sum+=a
    print(a,end=' ')
print()
print("sum of ele. of slice1:",sum)
print("Slice2")
sum=0
for a in slc2:
    sum+=a
    print(a,end=' ')
print("sum of ele. of slice2:",sum)
```

```
avg=sum/len(slice2)
print("average of ele. of slice2:",avg)
```

FILE : CH10EG3

```
l1=[17,24,15,30,34,27]
l2=l1.copy()
print("original list ",l1)
print("created copy of the list:",l2)
l2[0]+=10
l2[-1]+=10
print("copy of the list after change:",l2)
print("original list:",l1)
```

FILE : CH10EG3A

```
#1
print("#1")
lis=[13,18,11,16,18,14]
print(lis.index(18))
print(lis.append(33))      # may be asked as output
print(lis.index(33))
lst=lis    # a list can't be transfer to another list
lst=lis.append(12)
print(lst)
```

```
#2
print("#2")
t1=[1,3,5]
t2=[7,8]
t1.append(t2)
t1.append(10)
#t1.append(11,12)      # unable to append more than one item or
                        # element
t1.extend([11,12])     # in order to add more than one item than
                        # they must be enter as a list
print (t1)             # use .extend(...)
```

#3

```
print("#3")
t1.insert(2,33)    # SYNTAX    .insert(index,item)
print (t1)
t1.pop(2)          # SYNTAX    .pop(index) to delete an item at
index pos
print (t1)
t1.remove(11)      # SYNTAX    .remove(item) to delete an item
print (t1)
print(t1.count(18))    # may be asked as output
```

```
t1.reverse()      # to reverse the entire list
print (t1)        # may be asked as output (items in a list
```

with in a list)

```
t2=['e','i','q','a','q','p']
print (t2)
t2.sort()          # may also be written as
t2.sort(reverse=False)
print (t2)         # by default increasing order
```

```
t2.sort(reverse=True)
print(t2)
t2.sort(reverse=False)    # may be asked as output
print(t2)
```

```
print(max(t2))
print(min(t2))
```

FILE :

```
t1=[1,3,5]
print(sum(t1))    # add up all integers
sval=[17,24,15,30]
rsval.sorted(val,reverse=True)    # may also be written as
t2.sort(reverse=False)
print (rsval)    # by default increasing order
t1.clear()       # to clear the whole list
print (t1)
```

#4

```
print("#4")
l2=[[1,2,3],[5,6]]    # it is known as ragged
print(l2)
```

FILE : CH10EG4

```
lst=[]
r=int(input("how are rows?"))
c=int(input("how are columbs ?"))
for i in range(r):
    row=[]
    for j in range(c):
        elem=int(input("Element in row "+str(i+1)+" , col "+str(j+1)+" : "))
        row.append(elem)
    lst.append(row)
print("list created is:",lst)
print("-----")
print("list create is:")
print("lst=[")
for i in range(r):
    print("\t[",end=" ")
    for j in range(c):
        print(lst[i][j],end=" ")
    print("]")
print("\t]")
# difference between append and extend is that in append
```

FILE : CH10EG5

```
#1
print("#1")
val=[17,24,15,30]
print("The original list is : ",val)
print()
val.extend([34,27])
print("The extended list is : ",val)
```

```

#2
#print("#2")
print()
c=int(input("How many elements to be added to the list ?"))
for j in range(c):
    elem=int(input("Element "+str(j+1)+": "))
    val.extend([elem])
print()
print("The further extended list is : ",val)
#3
print("#3")
print()
c=int(input("How many elements to be added to the list ?"))
for j in range(c):
    pos=int(input("Enter element position : "))
    ele=int(input("Enter the element to be inserted/added at
position "+str(pos)+": "))
    val.insert(pos,ele)
print()
print("The changed list is : ",val)
#4
print("#4")
print()
c=int(input("How many elements to be deleted from the list ?"))
for j in range(c):
    pos=int(input("Enter element position : "))
    val.pop(pos-j)
print()
print("The changed list is : ",val)
print("the element at position 3 is ",val[3])
#5
print("#5")
print()
#d=int(input("How many elements to be deleted from the list ?"))
d=int(input("Enter the element to be deleted : "))
for j in range(len(val)):
    if d==val[j]:
        val.remove(d)
print("The changed list is : ",val)

```

FILE : CH10EG6

```
myl=[2,4,6]
print("Existing list is :",myl)
n=eval(input("enter a number or a list to be appended:"))
if type(n)==type([]):
    print("you have accepted a list element")
    myl.extend(n)
elif type(n)==type(1):      # 1 stands for int
    print("you have accepted a int")
    myl.append(n)
else:
    print(" Please enter either an integer or a list.")# for a
SET
print("appended list is:",myl)
```

FILE : CH10EG7

```
val=[17,23,18,19]
print("the list is:",val)
while True:
    print("main menu")
    print("1.insert")
    print("2.delete")
    print("3.exit")
    ch=int(input("Enter your choice1/2/3 "))
    if ch==1:
        item=int(input("Enter item "))
        pos=int(input("insert at which position "))
        index=pos-1
        val.insert(index,item)
        print("success! list now is: ",val)
    elif ch==2:
        print("delete menu ")
        print("1.delete using value ")
        print("2.delete using index ")
        print("3.delete a sublist ")
        dch=int(input("enter choice (1 or 2 or 3) "))
```

```

        if dch==1:
            item=int(input("enter item to be deleted : "))
            val.remove(item)
            print("list now is:",val)
        elif dch==2:
            index=int(input("Enter index of item to be deleted
"))
            val.pop(index)
            print("list now is:",val)
        elif dch==3:
            l=int(input("Enter lower limit of list to be deleted
"))
            h=int(input("Enter upper limit of list to be deleted
"))
            del val[l:h]
            print("list now is: ",val)
        else:
            print("valid choice are 1/2/3 only ")
    elif ch==3:
        break
    else:
        print("valid choice are 1/2/3 only")

```

FILE : CH10EG8

```

val=eval(input("enter a list: "))
print("original list :",val)
val=val*2
print("Replicated list:",val)
val.sort()
print("sorting in ascending order :",val)
val.sort(reverse=True)
print("sorting in descending order :",val)

```

FILE : CH10EG9

```
lst=eval(input("enter list:"))
length=len(lst)
min_ele=lst[0]
min_index=0
for i in range(1,length):
    if lst[i] < min_ele:
        min_ele=lst[i]
        min_index=i
print("Given list is:",lst)
print("the minimum element of the given list is:")
print(min_ele,"at index",min_index)
```

FILE : CH10EG10

```
lst=eval(input("Enter a list:"))
length=len(lst)
mean=sum=0
for i in range(0,length):
    sum+=lst[i]
mean=sum/length
print("given list is:",lst)
print("the mean of given list is :",mean)
```

FILE : CH10EG11

```
lst=eval(input("enter a list:"))
length=len(lst)
element=int(input("enter element to be searched for:"))
for i in range(0,length):
    if element == lst[i]:
        print(element,"found at index",i)
        break
else:
    print(element,"not found in given list")
```

FILE : CH10EG12

```
lst=eval(input("enter a list:"))
length=len(lst)
element=int(input("enter element to be searched for:"))
count=0
for i in range(0,length):
    if element == lst[i]:
        count+=1
if count ==0:
    print(element,"not found in given list")
else:
    print(element,"has frequency as",count,"in given list")
```

FILE : CH10EG13

```
lst=eval(input("enter list: "))
length=len(lst)
uniq=[]
dupl=[]
count=i=0
while i<length:
    element=lst[i]
    count=1
    if element not in uniq and element not in dupl:
        i+=1
        for j in range(i,length):
            if element==lst[j]:
                count+=1
        else:
            print("element",element,"frequency:",count)
            if count==1:
                uniq.append(element)
            else:
                dupl.append(element)
    else:
        i+=1
print("original list",lst)
print("unique element list",uniq)
print("duplicates element list",dupl)
```

FILE : CH10EG14

```
lst=eval(input("Enter a list:"))
ln=len(lst)
mx=max(lst)
ind=lst.index(mx)
if ind<(ln/2):
    print("The maximum element",mx,"lies in the 1st half")
elif ind==(ln/2):
    print("The maximum element",mx,"lies in the middle")
else:
    print("The maximum element",mx,"lies in the 2st half")
```

FILE : CH10EG15

```
lst1=eval(input("Enter list1:"))
lst2=eval(input("Enter list2:"))
mx1=max(lst1)
mx2=max(lst2)
if mx1>=mx2:
    print(mx1,"the maximum is a list1 at index",lst1.index(mx1))
else:
    print(mx2,"the maximum is a list2 at index",lst2.index(mx2))
```